7CS033 Assignment Task 4

Complete the following tasks.

You should comment on your code where appropriate. Failing to add comments may result in a mark deduction.

You should write any explainations in the report cell if the task explicitly asks for this or if you wish to expand upon your solution.

Ensure you have ran your code cell and printed any of the required outputs. If necessary you can use more than one code cell.

Tasks

TASK 4.1: Download the adult_T4 dataset. Apply K-Means and Hierarchical clustering to three optional columns in the dataset. Find the optimum number of clusters for both clustering methods (10%).

NOTE: You should comment on your code wherever necessary and briefly explain what the code is doing

In [103…

```
############# WRITE YOUR CODE IN THIS CELL (IF APPLICABLE)####################

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from sklearn.cluster import AgglomerativeClustering
from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.decomposition import PCA


from sklearn.preprocessing import MinMaxScaler
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import OrdinalEncoder

from mpl_toolkits.mplot3d import Axes3D
```

In [121…

```
df = pd.read_csv('adult_T4.csv')

d1 = df['age'] #target variable 1
d2 = df['education-num'] #target variable 2
d3 = df['hours-per-week'] #target variable 3
```

In [122…

```
# Select three numerical columns for clustering
# These variables are continuous and suitable for distance-based clustering

X = pd.concat([d1,d2,d3], axis = 1); #get our 3 input variables
#X = X.dropna()
```

In [123…

```
# Remove missing values (clustering cannot handle NaNs)
X = X.dropna()
```

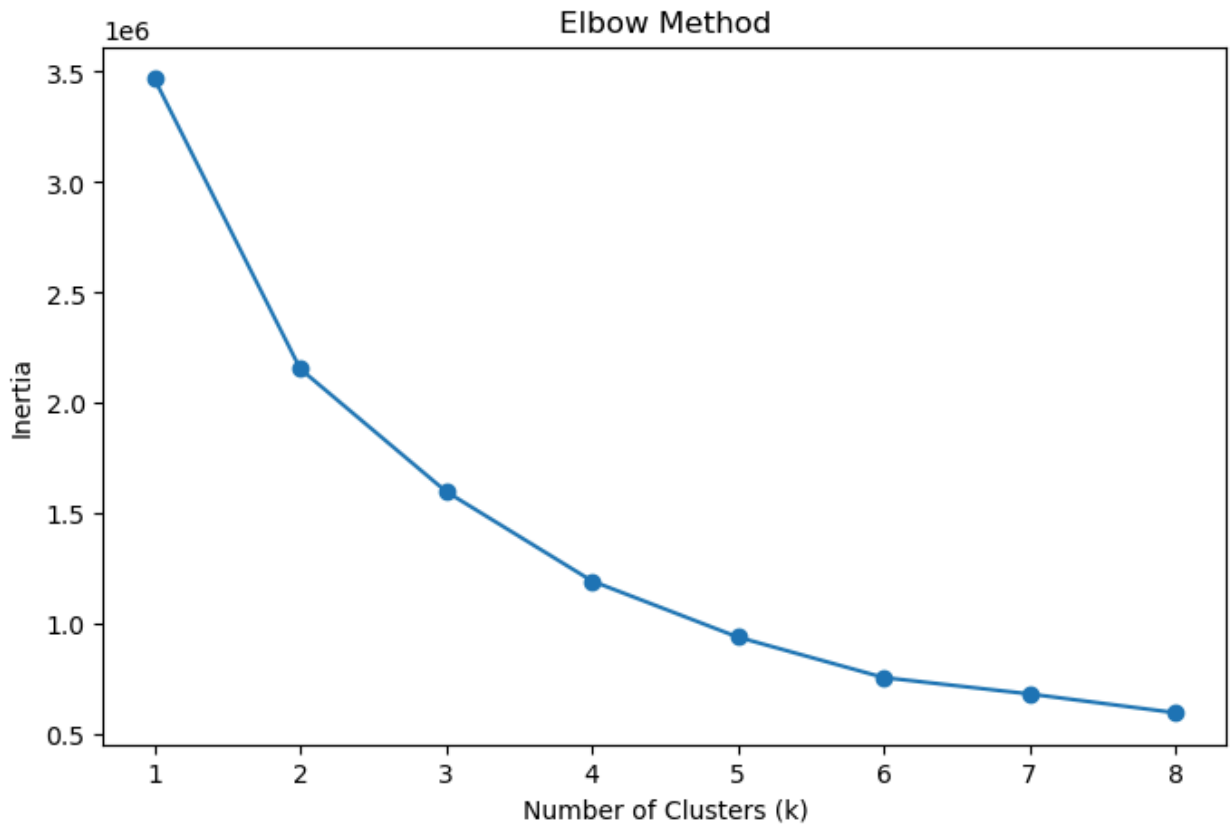
In [124…

```
# Scaling is important because clustering uses distance metrics
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

K-MEANS CLUSTERING

In [125…

```
# --- Elbow Method ---
# Used to identify the point where adding more clusters
# does not significantly reduce inertia

inertia = []


for k in range(1, 9):
    km = KMeans(n_clusters = k, random_state = 42)
    km.fit(X)
    inertia.append(km.inertia_)

#plot the elbow curve
plt.figure(figsize=(8,5))
plt.plot(range(1,9), inertia, marker = 'o')
plt.xlabel('Number of Clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method')
plt.show()
```

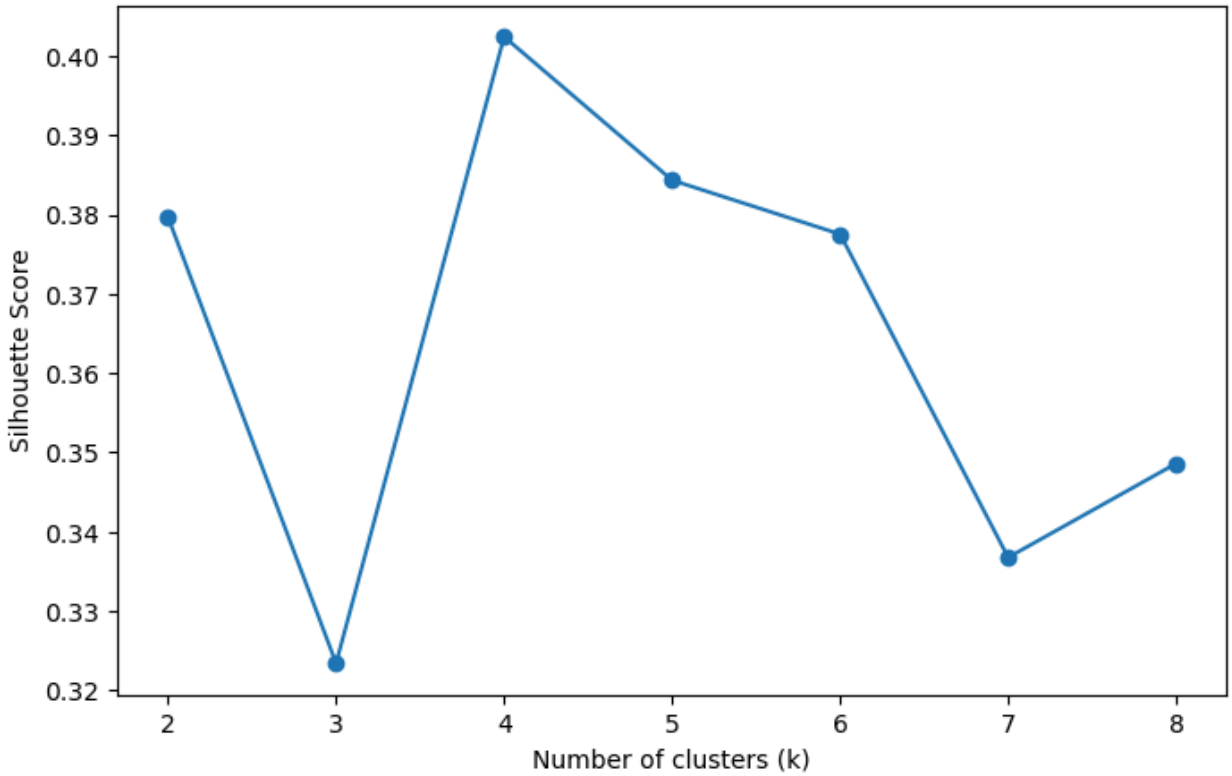


In [ ]:

In [126…

```
# Silhouette Method
# Measures how well points fit within their clusters

sil_scores = []

#
for k in range (2, 9):
    km = KMeans(n_clusters = k, random_state = 42)
    labels = km.fit_predict(X)
    score = silhouette_score(X, labels)
    sil_scores.append(silhouette_score(X, labels))
    print(f"k={k}, Silhouette Score={score}")


plt.figure(figsize = (8, 5))
plt.plot(range(2,9), sil_scores, marker = 'o')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Silhouette Score')
plt.show()
```

```
k=2, Silhouette Score=0.37965490443771743
k=3, Silhouette Score=0.323411684363288
k=4, Silhouette Score=0.4024432969339395
k=5, Silhouette Score=0.3843419106780839
k=6, Silhouette Score=0.3775365397877103
k=7, Silhouette Score=0.3367647194464343
k=8, Silhouette Score=0.3486205456416944
```



In [129…

```
# Optimal k is the one with highest silhouette score
optimal_k_kmeans = np.argmax(sil_scores) + 2
print("Optimal k for KMeans:", optimal_k_kmeans)
```

```
Optimal k for KMeans: 4
```

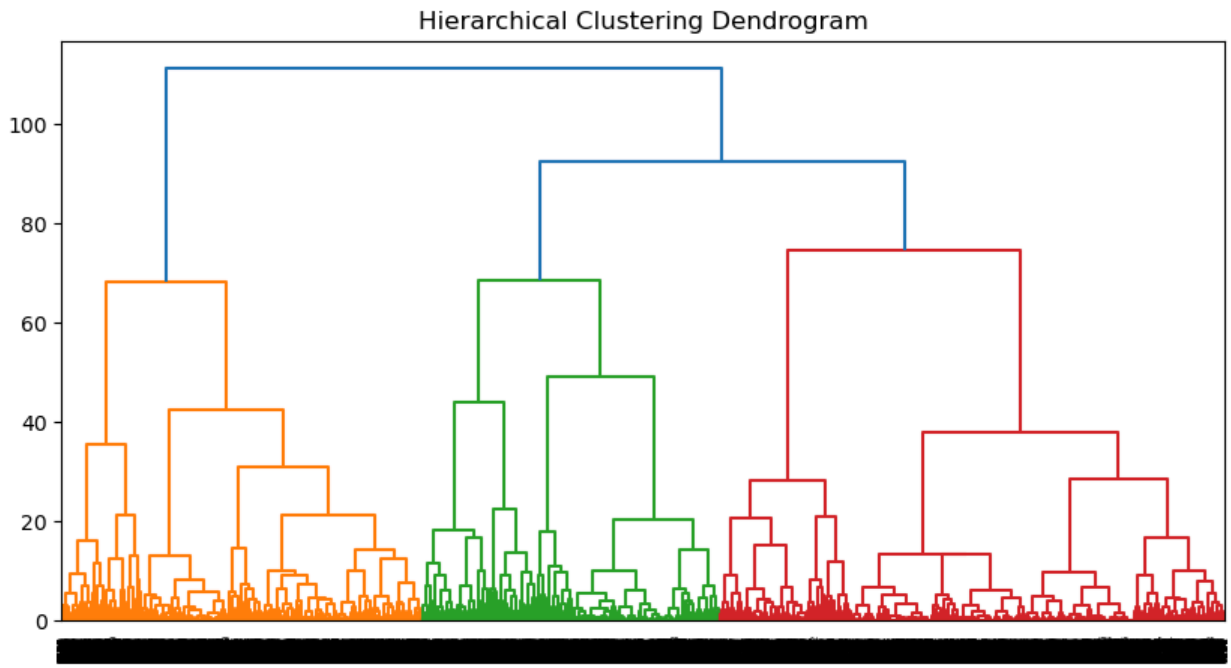
In [130…

```
# Fit final KMeans model
kmeans = KMeans(n_clusters=optimal_k_kmeans, random_state=42)
kmeans_labels = kmeans.fit_predict(X_scaled)
```

HIERARCHICAL CLUSTERING

In [131…

```
#  Dendrogram
# Visual representation of hierarchical cluster merging

linked = linkage(X_scaled, method='ward')

plt.figure(figsize=(10,5))
dendrogram(linked)
plt.title("Hierarchical Clustering Dendrogram")
plt.show()
```

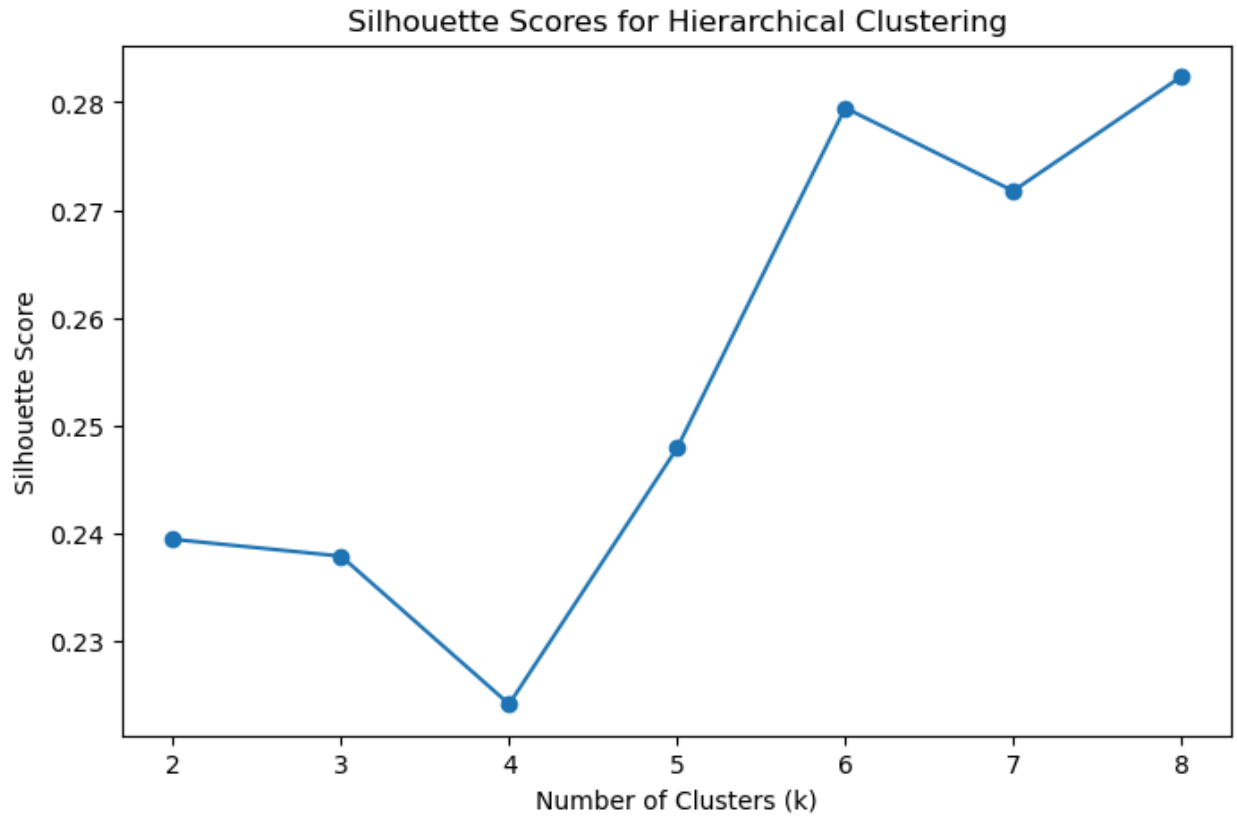


In [ ]:

In [133…

```
#  Silhouette Method for Hierarchical

hier_scores = []

for k in range(2, 9):
    agg = AgglomerativeClustering(n_clusters=k, linkage='ward')
    labels = agg.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    hier_scores.append(score)
    print(f"Hierarchical k={k}, Silhouette Score={score:.4f}")

plt.figure(figsize=(8,5))
plt.plot(range(2, 9), hier_scores, marker='o')
plt.xlabel("Number of Clusters (k)")
plt.ylabel("Silhouette Score")
plt.title("Silhouette Scores for Hierarchical Clustering")
plt.show()
```

```
Hierarchical k=2, Silhouette Score=0.2395
Hierarchical k=3, Silhouette Score=0.2379
Hierarchical k=4, Silhouette Score=0.2242
Hierarchical k=5, Silhouette Score=0.2479
Hierarchical k=6, Silhouette Score=0.2795
Hierarchical k=7, Silhouette Score=0.2718
Hierarchical k=8, Silhouette Score=0.2824
```



In [ ]:

In [135…

```
optimal_k_hier = np.argmax(hier_scores) + 2
print("Optimal k for Hierarchical:", optimal_k_hier)
```

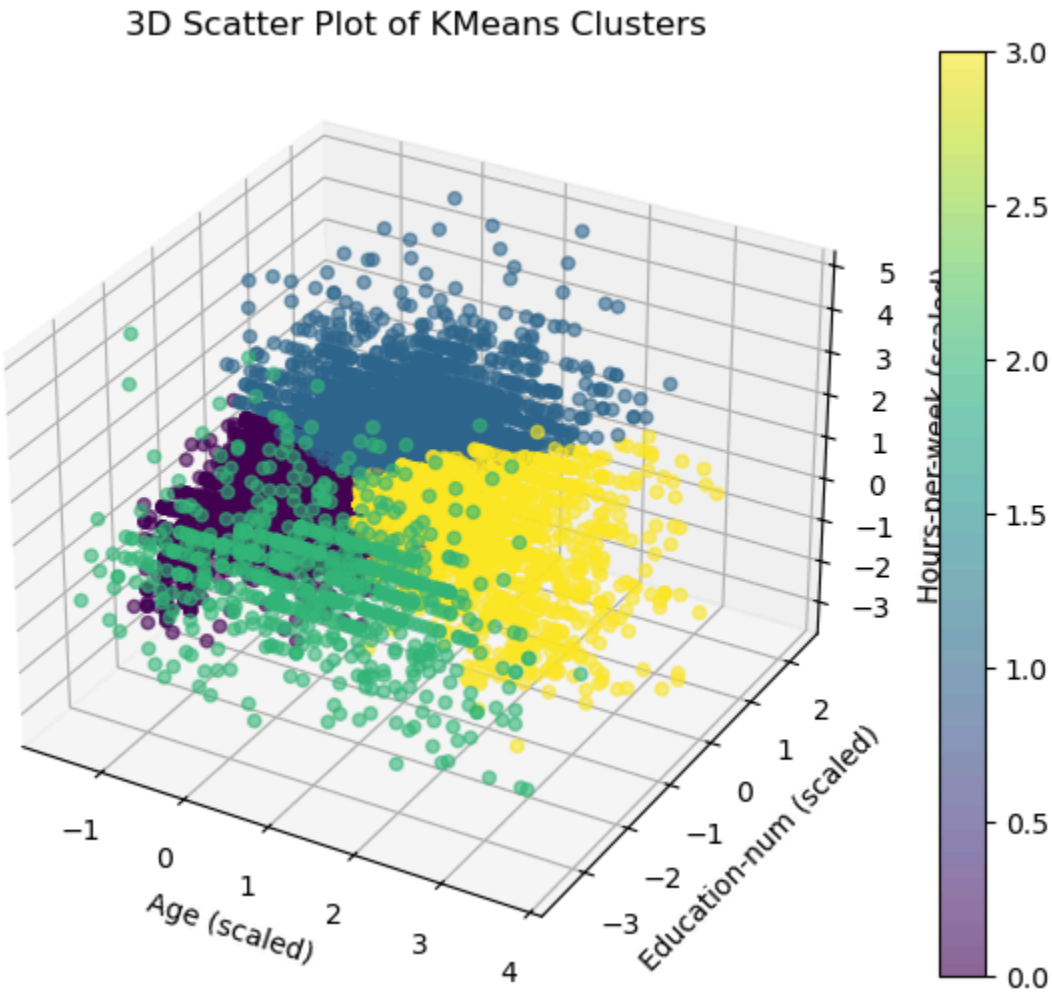
```
Optimal k for Hierarchical: 8
```

3D Scatter Plot for KMeans

In [149…

```
# Create 3D figure
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

centroids = kmeans.cluster_centers_


# Plot clusters
scatter = ax.scatter(
    X_scaled[:, 0],   # age
    X_scaled[:, 1],   # education-num
    X_scaled[:, 2],   # hours-per-week
    c=kmeans_labels,
    cmap='viridis',
    alpha=0.6
)

# Axis labels
```

```
ax.set_xlabel("Age (scaled)")
ax.set_ylabel("Education-num (scaled)")
ax.set_zlabel("Hours-per-week (scaled)")
ax.set_title("3D Scatter Plot of KMeans Clusters")

plt.colorbar(scatter)
plt.show()
```



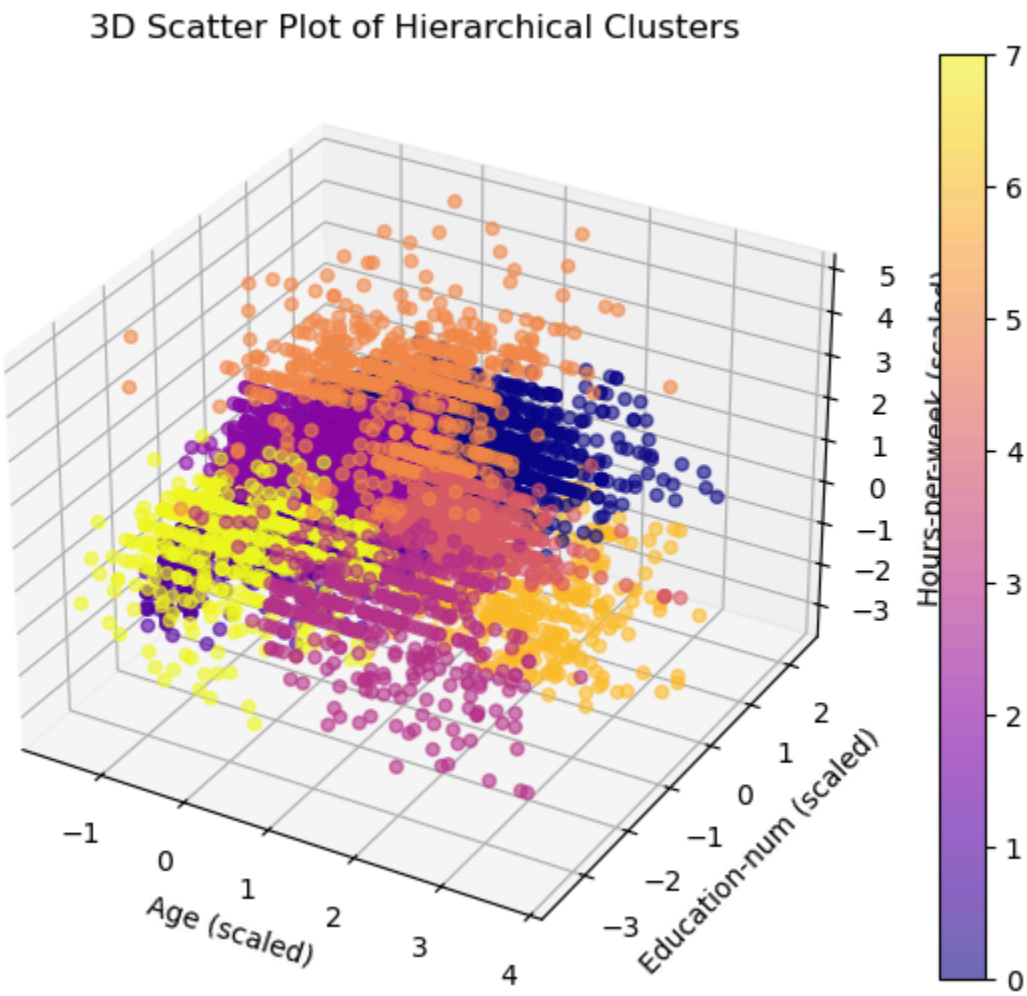
In [ ]:

3D Plot for Hierarchical Clustering

In [147…

```
# Fit hierarchical model using optimal k
agg = AgglomerativeClustering(n_clusters=optimal_k_hier)
hier_labels = agg.fit_predict(X_scaled)

fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')

scatter = ax.scatter(
    X_scaled[:, 0],
```

```
    X_scaled[:, 1],
    X_scaled[:, 2],
    c=hier_labels,
    cmap='plasma',
    alpha=0.6
)

ax.set_xlabel("Age (scaled)")
ax.set_ylabel("Education-num (scaled)")
ax.set_zlabel("Hours-per-week (scaled)")
ax.set_title("3D Scatter Plot of Hierarchical Clusters")

plt.colorbar(scatter)
plt.show()
```



In [ ]:

############# WRITE YOUR REPORT IN THIS CELL (IF APPLICABLE)################### In this task, K-Means and Hierarchical clustering were applied to three numerical features: age, education-num, and hours-per-week. The data was standardised to ensure equal contribution of each feature to the distance calculations. The Elbow method and Silhouette score were used to determine the optimal number of clusters. For K-Means, the highest silhouette score was obtained at k = 4, indicating that four clusters provide the best separation and cohesion. For Hierarchical clustering, the highest silhouette score was observed at k = 8, although k = 4 produced a similar value. Overall, K-Means slightly outperformed Hierarchical clustering based on

silhouette score, suggesting clearer cluster separation in the selected feature space. To further examine cluster structure, 3D scatter plots were generated using the three selected features. The visualisations show moderate separation between clusters, although some overlap exists, which is consistent with the moderate silhouette scores. This suggests that while meaningful cluster structure is present, the boundaries between clusters are not sharply defined.

Task 4.2: Apply the PCA method to the dataset and extract the first two principal components (n_components=2). Plot the scatter plot of the dataset's first two components for the two classes of the income column (5%).

NOTE 1: You should comment on your code wherever necessary and briefly explain what the code is doing.

NOTE 2: You need to encode the categorical columns, normalise the dataset, and remove the income column before applying the PCA method.

In [118…

In [138…

```
# Replace '?' values with NaN
df.replace("?", np.nan, inplace=True)

# Drop rows with missing values
df2 = df.dropna()
```

In [ ]:

In [139…

```
#Encode Categorical Variables
encoder = OrdinalEncoder()
categorical_cols = df2.select_dtypes(include='object').columns

df2[categorical_cols] = encoder.fit_transform(df2[categorical_cols])
```

```
/var/folders/6b/x27jrjp92kn9r5_84c_ywfhcsgm8y7/T/ipykernel_55423/1630149546.p
y:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/sta
ble/user_guide/indexing.html#returning-a-view-versus-a-copy
  df2[categorical_cols] = encoder.fit_transform(df2[categorical_cols])
```

In [140…

```
# Separate features and target variable
```

```
X2 = df2.drop('income', axis=1)
y2 = df2['income']
```

In [ ]:

In [143…

```
#Normalise the Dataset

# PCA is affected by scale, so we normalise
scaler = MinMaxScaler()
X2_scaled = scaler.fit_transform(X2)
```

Apply PCA

In [144…

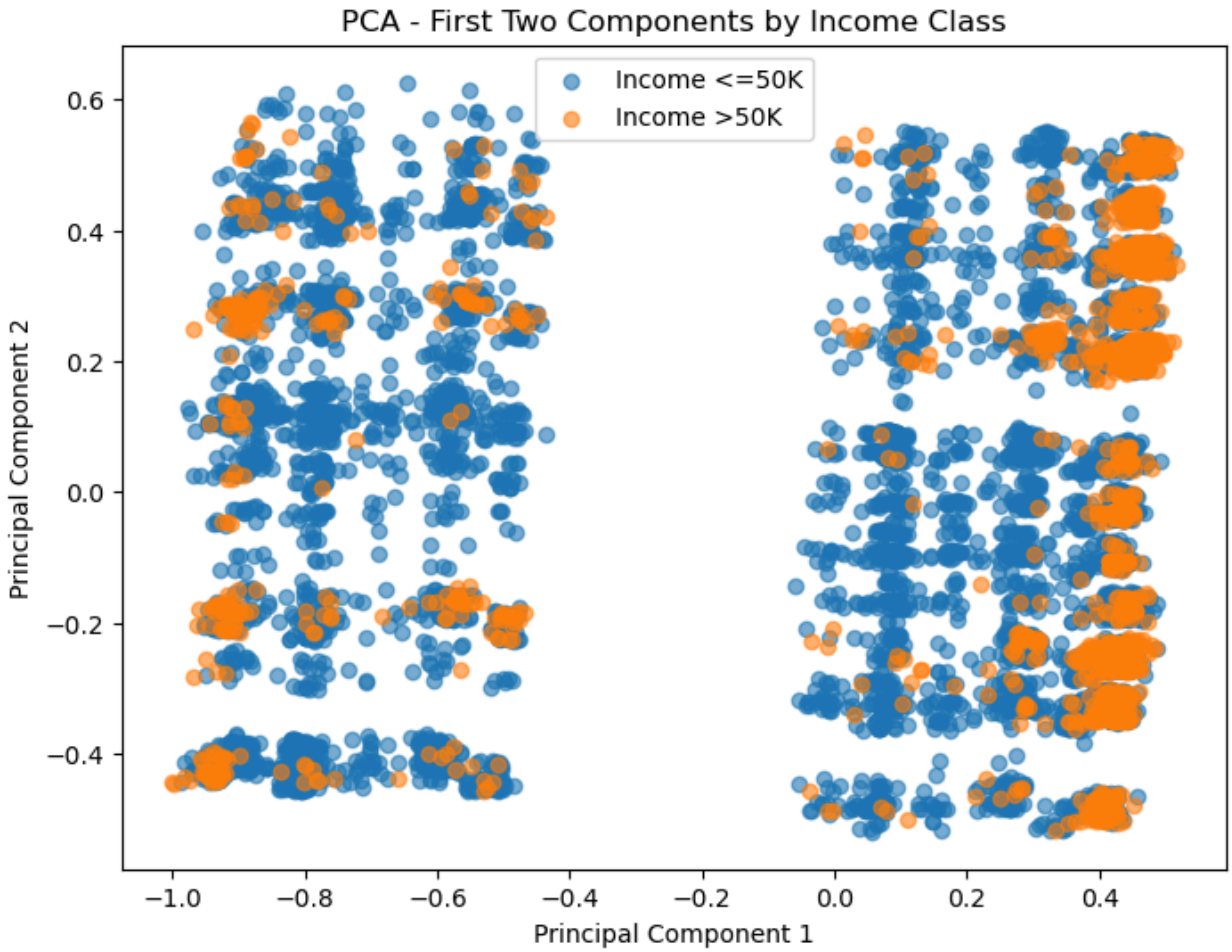
```
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X2_scaled)

print("Explained Variance Ratio:", pca.explained_variance_ratio_)
```

```
Explained Variance Ratio: [0.37235078 0.12949804]
```

In [ ]:

In [145…

```
#Plot PCA for the Two Income Classes
# Create masks for the two income classes
class_0 = y2 == 0
class_1 = y2 == 1

plt.figure(figsize=(8,6))

# Plot each class separately
plt.scatter(X_pca[class_0, 0], X_pca[class_0, 1], label="Income <=50K", alpha=

plt.scatter(X_pca[class_1, 0], X_pca[class_1, 1], label="Income >50K", alpha=0

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - First Two Components by Income Class")
plt.legend()
plt.show()
```



############# WRITE YOUR REPORT IN THIS CELL (IF APPLICABLE)################### Before applying PCA, categorical variables were encoded and the dataset was normalised. The income column was removed prior to dimensionality reduction. The first two principal components explained approximately 50% of the total variance in the dataset. A scatter plot of these components, coloured by income class (≤50K and >50K), showed noticeable overlap between the two classes. This suggests that income is not clearly separable in two-dimensional space and may depend on more complex feature interactions.

In [ ]: